FACULTY
OF INFORMATION
TECHNOLOGY
CTU IN PRAGUE

Assignment of bachelor's thesis

Title:	StudySpace: A Collaborative Web Application for Academic Students
Student:	Thet Oo Aung
Supervisor:	Ing. Jan Blizničenko, Ph.D.
Study program:	Informatics
Branch / specialization:	Software Engineering 2021
Department:	Department of Software Engineering
Validity:	until the end of summer semester 2026/2027

Instructions

The goal of this thesis is to design and implement a functional prototype of a web application that combines course materials management and student communication into a single environment.
The application aims to reduce the need for switching between different tools by linking study materials directly to peer discussions.

Key Functional Requirements:
1. Course Administration Module: An interface for instructors to create course structures, upload lecture materials, and handle student enrollments.
2. Content Extension System: This feature allows students to clone course materials into private workspaces for personal note-taking and group collaboration, or to propose their modifications to the instructor. The instructor can manually review them and choose to showcase the student's extended version alongside the official content for the entire class.
3. Contextual Messaging System: A communication tool that allows students to "anchor" questions to specific course materials. By typing a shortcut symbol (such as @), students can select a PDF or files from the course materials to include in their message. This creates a clickable link, allowing others to jump directly to the selected document for quick reference and support.
4. AI-Based Content Querying: Integration of the APIs from Large Language Models (e.g. OpenAI, Gemini) to allow users to query summaries or explanations based on the

context of the loaded study materials.
5. Access Control: A secure authentication/ authorization system to differentiate permissions depending on the roles: eg. Students (view/copy/discuss) and Instructors (manage content/approve merges).

Follow these steps:
1. Analyze Requirements: Define the functional requirements for StudySpace and analyze the specific needs of the target academic audience through user research.
2. Current Solutions Analysis: Research existing solutions (Learning Management Systems like Moodle, communication tools like Discord/Teams) to identify integration gaps and opportunities for improvement.
3. UI Design: Create wireframes to visualize the prototype's flow for a better user experience.
4. System Architecture: Design the database schema and API structure. Research and select the appropriate technology stack (programming languages, frameworks, libraries, tools), specifically addressing the data model required for personal content copies, merge proposals and contextual message anchoring.
5. Implementation: Prioritize the development of a functional prototype of the application with the frontend and backend components, integration of the LLM API, and implementation of all core features.
6. Testing: Perform functional and usability testing to ensure system stability and completeness of the core features.
7. Documentation: Write technical documentation for deployment and user guides for both students and instructors.